

ARINC 429 Data Bus Project Proposal

Caterpillar

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I. PROTOCOL AND MISSION NEED

A. *Domain and Mission Need*

The domain that this paper will be on, is in the Aviation domain, and the protocol that we are going to be focusing on is the ARINC 429 data bus. Our paper will plan to address the threat vectors to the data bus and resolve them by implementing a filter of data to the data bus so that any information that is sent is validated to be from the plane and not outside of the plane.

B. *Justification*

This protocol is paramount because it handles all the communication within the plane and any of the data being transmitted across the plane. Whether that be the flight information at that moment to the black box, or the radar data that is being displayed could not be visible on the screen in the cockpit.

II. INITIAL FEASIBILITY AND RISK ASSESSMENT

A. *Complexity*

Our group does not believe that this project is very complex since it is mainly hardware that we are dealing with and no software. The documentation is primarily about how the data is formatted and read throughout the plane.

B. *Project Risks*

The biggest risk to our project is that we may not be capable of simulating an attack on the data bus and if we have the capacity to do achieve our goal, it may not be what we sought out. The only mitigation that we anticipate would be simulating an attack on systems that use the data bus instead of the physical bus.

III. STAKEHOLDER

A. *Identifying*

There are a plethora of stakeholders that would be invested into this paper, such as pilots, commercial aviation companies, the air traffic control tower, airport operators, passengers or anyone who flies on a commercial plane, technicians, the stakeholders of the commercial companies, manufacturers, and attackers, like insiders, nation-states, and terrorists. All of the people that would want to take note of the paper, besides the attackers, all have a stake in the company, whether it be their job/livelihood, money that they put into the company (stocks), and/or their physical life because if the plane goes down then those passengers lives will be heavily affected.

IV. THREAT ANALYSIS

A. *Spoofing*

An attacker can spoof the sending or receiving system of the data bus through physical access or a system specific method.

B. *Tampering*

An attacker who has successfully spoofed the sending entity on the data bus, or an attacker who has managed to tap the wire, can alter the data sent over the bus allowing for greater control of the plane's systems.

C. Repudiation

Currently there is no non-repudiation in the airbus system, allowing for attacks to go unmonitored.

D. Information Disclosure

An attacker who successfully spoofed the receiving end of the data bus or tapped the wire can read packets sent over the data bus and gain potentially sensitive flight information.

E. Denial Of Service

An attacker who gains access to send messages over the data bus can send many null or junk packets over the data bus potentially creating a bottleneck.

F. Escalation of Privilege

An attacker who has the ability to tamper with the frames sent over the bus can alter frames 9 and 10, these bits control what system the frame is being sent to, allowing the attacker to send a frame to a system with higher privilege than originally intended.

V. PRELIMINARY SYSTEM ENGINEERING MODELS

A. Scope

The Scope of the project includes simulating the databus by sending correctly formatted data from one system to another. Simulating an attack on an identified "Vulnerable system" or assuming an attack on the airbus via physical access, insider threat, or supply chain, and using the databus to send malicious messages to another system on the plane. The hardening process will include applying a filtering technique to the data bus to allow it to ignore potentially malicious frames sent over the bus.

B. SysML Diagram

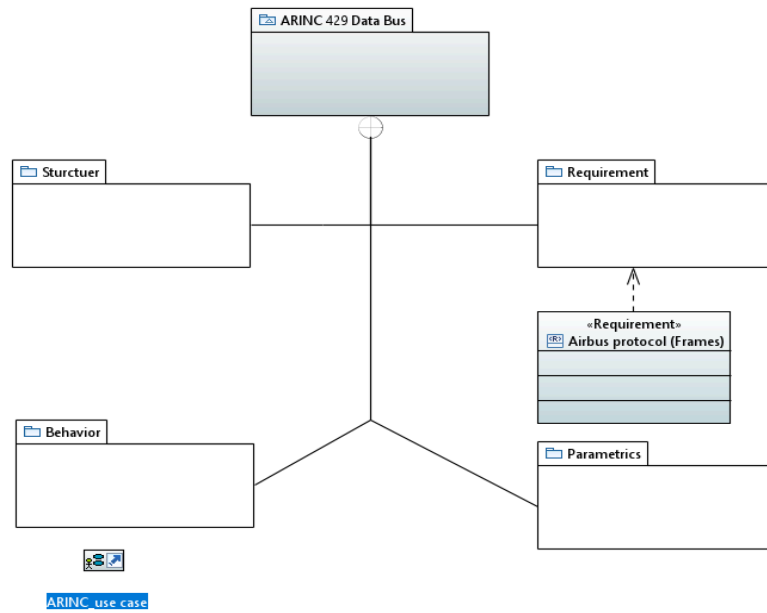


Figure 1: Package Diagram

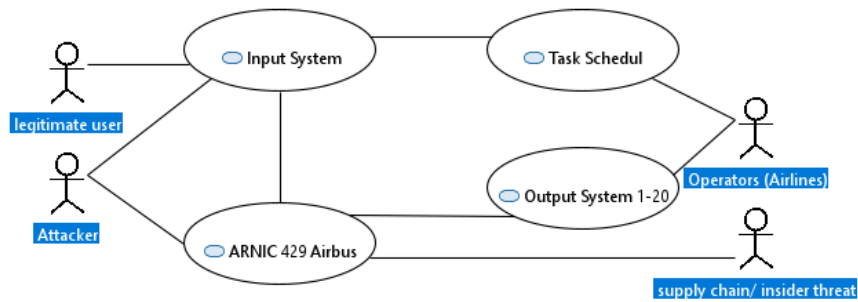


Figure 2: Use Case Diagram

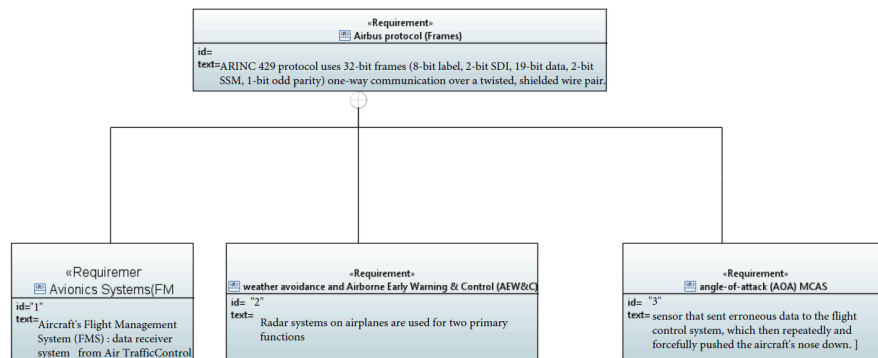


Figure 3: Requirements Diagram

The general use case for the ARNIC 429 airbus, according to Figure 2, is a middleman for data transfer internally in the plane. In an interaction with the airbus, it starts with a user or a task scheduler sending data to an input system, after that the input system sends the data through the airbus and the airbus sends it to one of twenty output systems. It is important to note input and output systems can be any system on the plane including other airbus's. We have identified that an attacker will either attack the data before it gets to the airbus at the input system, or if the attacker has physical access or successfully infiltrates the airbus supply chain they can directly attack the data in transit over the airbus.

C. Requirements

The requirements for the ARNIC 429 airbus protocol requires that all data, called frames or words, is formatted in 32 bit and consists of an 8 bit label that contains context for the data (e.g. elevation data, velocity data, radar data), a 2 bit SDI that contains information about the receiving data sink, 19 bits of data, a 2 bit SSM that contains additional information depending on the label (ex negative or positive, cardinal direction), and a 1 bit P value that ensures the total number of ones is odd if there is no error in transmission, or even if there is an error. Additionally the protocol requires all airbus's to be a twisted and shielded wire pair that only allows for one way communications.

VI. PROJECT MANAGEMENT PLAN

A. Team Roles

Currently, we are all researching attack vectors and trying to figure out what vector is the most probable. It is expected that everyone at least comes up 1-2 potential attack vectors. But, as our roles stand we will take it week by week and each group member will choose a task to work on.

B. GitHub Repository

We have a Git Repository with our instructor added as a collaborator as well as all of our group members.

<https://github.com/JoshuaMFrench/CYSE465-Caterpillars-Group/>

VII. REFERENCES

- IEEE Reference Guide, IEEE Publication Operations, 445 Hoes Lane, Piscataway, NJ, November 29th, 2023. [Online]. Available: https://journals.ieeeauthorcenter.ieee.org/wp-content/uploads/sites/7/IEEE_Reference_Guide.pdf
- B. Waldmann. *ARNIC 429 Tutorial*, v2.2. (2019). Accessed: September 12th, 2025. [Online]. Available: <https://www.aim-online.com/wp-content/uploads/2019/07/aim-tutorial-oview429-190712-u.pdf>

Project Plan